MKALAMA DISTRICT COUNCIL
FORM TWO MOCK EXAMINATION
BASIC MATHEMATICS

MARKING SCHEME

1. (a). Amina, Juma and Ally are training themselves about how to lap in their music. Amina, laps once every 9 minutes, Juma laps once every 15 minutes, and Ally laps every after 20 minutes. How long will it take for them to lap together

2	9	15	20
2	9	15	10
3	9	15	5
3	3	5	5
5	1	5	5
	1	1	1

solution

wazaelimu.com

(2 marks)

Time of lapping together= 2x2x3x3x5= 180 minutes= 3 hours *(3 marks)*

(b). Given; $17816273\ kg$ of cotton

i. $17816273 \approx 18000000$ (in millions)…… $02\frac{1}{2} marks$

ii. $17816273 \approx 17816000$ (in thousands)…….

$02\frac{1}{2}\ marks$

2a) If he spent $\frac{5}{9}$ of his income, then he remained with $1\ \frac{-5}{9} = \frac{4}{9}$ of Geofrey income is 8.8200 Tsh

$\frac{5}{9}$ of Geofrey income is (01 mark)

$= (\frac{5}{9}$ x Tsh 88200) $\div \frac{4}{9}$ …….. (01 mark)

$= \frac{5}{9}$ X Tsh 88200 x $\frac{9}{4}$ =110,250

Tsh 110,250 (01 mark)
Thus, he spent 110,250 Tsh .
His income = Income spent + the amount remained
= 110,250 + 88200 (01 mark)
= 198,450
Geofrey income was 198,450 (01 mark)

Alternativelly:
Total fraction is 1… 01mark
He remain with 1- 5/9 = 4/9… 01mark

Let Y be total income… 01mark

4/9 x Y=88200 01

Y=(88200X9)/4

=198,450/=… 01mark

2b) required first to find the LCM of denominators first as follows

… (01 marks)

2	24	72
2	12	36
2	6	18
3	3	9
3	1	3
	1	1

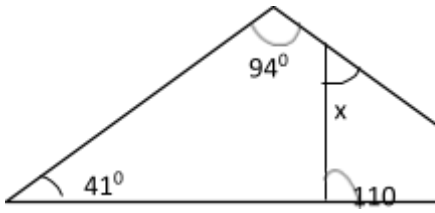
$LCM = 2 \times 2 \times 2 \times 3 \times 3 = 72$……..$(01 marks)$

Now; $\frac{15}{24} \times 72 =$ ……$\left(0\frac{1}{2} marks\right)$

$\frac{7}{72} \times 72 = 7$..... $\left(0\frac{1}{2} marks\right)$

Therefore; $\frac{15}{24}$ is greater than $\frac{7}{24}$ **……02*marks***

3(a). Given; $\Delta\ below,$



……$41^0 + 94^0 + y = 180^0$

$01 marks$

$135^0 + y = 180^0$

$y = 180^0 - 135^0$

…… $y = 45^0$

Also; $x + y + 110^0 = 180^0\ but\ y = 45^0$……$01\ marks$

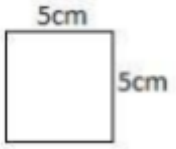
$x + 45^0 + 110 = 180^0$

$x + 155^0 = 180^0,$ $x = 180^0 - 155^0$

$x = 25^0$ ..... $0\frac{1}{2} marks$

Therefore; $x = 25^0$ and $y = 45^0$ 02*marks*

(b). Give; the square with length $5m$



..... $0\frac{1}{2} marks$

i). Perimeter $= 2(l + l) = 2(5 + 5)$

$= 2 \times 10 = 20m$

Therefore; Perimeter is $20m$.....02*marks*

ii). Area $l \times l = 5m \times 5m = 25m^2$

therefore; the Area is $25m^2$ .....02$\frac{1}{2}$*marks*

4) (a) $672^2 - 328^2$

From $a^{2-}b^2 =$ (a+b) (a-b).....

(01mark) $672^2 - 328^2 = (672 + 328)(672 - 328)$

$= (100)(344)$ (01 marks)

$672^2 - 328^2 = 344{,}000$ (01 marks)

(b) M*N = $\frac{M-n}{m+n}$

3*2 = M=3,n=2 (01 mark)

3*2 = $\frac{3-2}{3+2}$ … (01 mark)

3*2 = $\frac{1}{5}$ … (02 mark)

(c) Required to solve by general formula $-2y^2 + 7y - 6 = 0$

$a = -2, b = 7, c = -6$ .....(01 mark)

y= $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$ ..... (01 marks) y= $\frac{-7 \pm \sqrt{(7)^2-4(-2)(-6)}}{2(-2)}$

y= $\frac{-7 \pm \sqrt{49-48}}{-4}$

y= $\frac{-7 \pm \sqrt{1}}{2-4}$

y= $\frac{-7\pm1}{-4}$ (01/2 mark)

y= $\frac{-7+1}{-4}$ or y= $\frac{-7-1}{-4}$

y= $\frac{-6}{-4}$ or y= $\frac{-8}{-4}$

y= $^{3}/_{2}$ or y = 2 (001/2 marks)

5a) if a:b= 2:3 and b:c=5:6 find value of a:c

$\frac{a}{b} \times \frac{b}{c} = \frac{a}{c} = \frac{2}{3} x \frac{5}{6}$ *(2 marks)*

$= \frac{5}{9}$ *(2 marks)*

Hence a:c=5:9 *(1 mark)*

b) a shopkeeper bought a radio for Tsh 80,000 and sold it a profit of 20%. What were the profit and selling price?

Solution

Let X be selling price

80000 = 100 % *(2 marks)*

X = 120 %

Selling price= $80000 x \frac{120\%}{100\%}$=Tsh 96,000 *(1 mark)*

Profit = selling price – buying price *(1 mark)*

= 96,000 - 80,0000= Tshs 16,000 *(1 mark)*

6(a). Give; Point $P(3,1)$

i. Required to find image reflected in

x-axis From; $M_{x-axis}(x,y) = (x,-y)$

Now; $P_{x-axis}(3,1) = P^1(3,-1)$

Therefore; the image is $P^1(3,-1)$ ..... $02\frac{1}{2}$ marks

ii. Required to find image in line

$M_{y=x}(x,y) = (y,x)$

$P_{y=x}(3,1) = P^1(1,3)$

$y = x$ From;

Now,

Therefore; the image in line $y = x$ is $P^1(1, 3)$……..$02\frac{1}{2}marks$

(b). Given; $P(-4,5)$ and $Q(-2,1)$

Required to find equation in form of $y = mx + c$

From; $Gradient = \frac{y_2 - y_1}{x_2 - x_1}$ ..... $0\frac{1}{2} marks$

Now; $m = \frac{1-5}{-2-(-4)} = \frac{-4}{-2+4}$ .....$0\frac{1}{2} marks$

$m = \frac{-4}{2}, \quad m = -2$.....$01 marks$

Now, take any one point i.e $(-4,5)$ to get the equation

From; $y - y_1 = m(x - x_1)$ .....

$0\frac{1}{2} marks$

$y - 5 = -2(x - (-4)), \quad y - 5 = -2(x + 4)$ ..... $0\frac{1}{2} marks$

$y - 5 = -2x - 8, \quad y = -2x - 8 + 5$

$y = -2x - 3$ .....$01 marks$

Therefore; the equation is $y = -2x - 3$ .....$01 marks$

7 a) solve for x in $\left(\frac{3}{5}\right)^{x-6} = \left(\frac{25}{9}\right)^{x}$

Solution

$\left(\frac{3}{5}\right)^{x-6} = \left(\frac{5}{3}\right)^{2x}$ *(1 mark)*

$\left(\frac{3}{5}\right)^{x-6} = \left(\frac{3}{5}\right)^{-2x}$ *(1 mark)*

x-6=-2x *(1 mark)*

x+2x=6 *(1 mark)*

3x=6

X=2 *(1 mark)*

b). i). Given; $log3 = 0.4771, \ log5 = 0.6990$

Required to find $log45$

Now; $log45 = \log(9 \times 5)$ ……$0\frac{1}{2} marks$

$= log9 + log5$

$= log3^2 + log5$

$= 2log3 + log5$ ……$0\frac{1}{2} marks$

$= 2(0.4771) + 0.6990$

$= 0.9542 + 0.6990$

$= 1.6532$

Therefore; $log45 = 1.6532$ ……$01 marks$

ii). Given; Required to rationalize the denominator $\frac{\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

now; R.F

$\frac{\sqrt{2}}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{\sqrt{6}-\sqrt{4}}{\sqrt{3}(\sqrt{3}-\sqrt{2})+\sqrt{2}(\sqrt{3}-\sqrt{2})}$ …… $0\frac{1}{2} marks$

$= \frac{\sqrt{6}-2}{\sqrt{9}-\sqrt{6}+\sqrt{6}-\sqrt{4}}$ ……$0\frac{1}{2} marks$

$= \frac{\sqrt{6}-2}{3-2} = \frac{\sqrt{6}-2}{1} = \sqrt{6} - 2$

Therefore; $\frac{\sqrt{2}}{\sqrt{3}+\sqrt{2}} = \sqrt{6} - 2$ ……$01\frac{1}{2} marks$

8(a) Required to Find the solution of $|2x + 1| > 3$ and represent it on a number line

It follows that, $\pm(2x + 1) > 3$ (1mark)

Thus either $+(2x + 1) > 3 \; or \; -(2x + 1) > 3$ (1mark)

$2x + 1 > 3 \; or \; -2x - 1 > 3$

$2x > 2 \quad or \quad -2x > 4$

$x > 1 \quad or \quad x < -2$

Therefore the solution of the inequality is $x > 1 \; or \; x < -2$ (1mark)

- Number line; a student required to show the obtained solution above on a number line (2mark)

b) T = $2\pi\sqrt{\frac{l}{g}}$

$\frac{T}{2\pi} = \frac{2\pi}{2\pi}\sqrt{\frac{l}{g}}$ … (01 mark)

$\left(\frac{T}{2\pi}\right)^2 = \left(\frac{L}{g}\right)_2$ (01 mark

$\frac{T2}{4\pi2} = \frac{L}{g}$ …… (01 marks)

$\frac{4\pi^2 L}{4\pi^2} = \frac{T2_g}{4\pi^2}$… (01 marks)

L = $\frac{T2_g}{4\pi^2}$ (01 marks)

9(a) $2\sin 60^0 + \cos 30^0$

$\text{Sin}60^0 = \frac{\sqrt{3}}{2}$ … (01 marks)

$\text{Cos } 30^0 = \frac{\sqrt{3}}{2}$ … (01 mark)

$2\sin 60^0 + \cos 30^0 = 2x \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$ (01 mark)

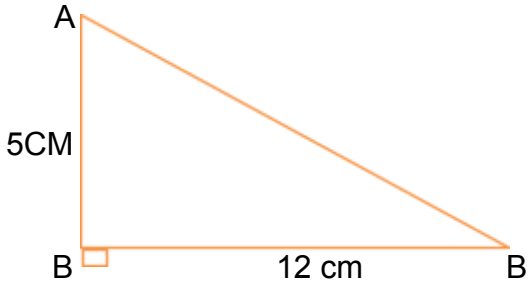
$= \sqrt{3} + \frac{\sqrt{3}}{2}$

$= \frac{2\sqrt{3}+\sqrt{3}}{2}$

$= \frac{(2+1)\sqrt{3}}{2}$ … (01 mark)

$= \frac{3\sqrt{3}}{2}$ … (01 mark)

(b)



(1 mark)

By using Pythagoras theory

$(\overline{AB})^2 = \overline{AC}^2 + (\overline{BC})^2$ (01 mark)

$= 5^2 + 122$

$25 + 144$ (01 mark)

$\sqrt{(\overline{AB})2} = \sqrt{169} = 13$ (02 marks)

The length of the hypotenuse is 13 cm.

10a) Required to solve the following simultaneous eqn by substitution method

$$\{\begin{matrix} 3x - 2y = 8 \ldots\ldots\ldots (i) \\ 2x + 5y = -1 \ldots\ldots\ldots (ii) \end{matrix}\}$$

From eqn make x the subject we get

$x = \frac{8+2y}{3} \ldots\ldots\ldots (iii)$ **(1mark)**

On substituting eqn …(iii) into eqn ….(ii) we get

$2\left(\frac{8+2y}{3}\right) + 5y = -1$ **(1mark)**

$16 + 4y + 15y = -3$

$19y = -19$

Then $y = -1$ **(1mark)**

Substitute the value of y into eqn..(iii)

$$x = \frac{8+2(-1)}{3} = \frac{8-2}{3} = 2$$

Therefore $x = 2 \; and \; y = -1$ **(2marks)**

(b) Required to find Amina's age

Let Amina's age = x **(1mark)**

John's age = x+6 **(1mark)**

Then, $x + x + 6 = 30$ **(1mark)**

$$2x + 6 = 30$$

$$2x = 24$$

$$x = 12$$ **(1mark)**

Therefore Amina's age =12 years